Updates Since Preliminary Result: BDT, NPB Removal, Double Interactions, and Shower Shape Comparisons

PPG12 Analysis Team (automated report)

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Abstract

This report summarizes the major analysis updates implemented since the preliminary isolated photon cross-section result in $p+p$ collisions at $\sqrt{s} = 200$ GeV with sPHENIX Run 24 data. Four work streams are covered: (1) the photon identification BDT, expanded from a minimal feature set to 11 model variants with an E_{T}-dependent parametric threshold; (2) non-physics background (NPB) rejection, upgraded from a hard timing cut to a dedicated 25-feature XGBoost score; (3) double-interaction efficiency studies, demonstrating that the apparent reconstruction efficiency drop in pileup MC is a truth–reco matching artifact; and (4) shower shape data–MC comparisons using pileup-blended MC templates, which improve χ^2/ndf by factors of up to 22×. Each section documents the methodology, key results, and implications for the cross-section measurement.

Contents

1 Introduction

The PPG12 isolated photon cross-section measurement uses sPHENIX Run 24 p+p data at $\sqrt{s} = 200$ GeV with an integrated luminosity of 16.6 pb^{-1} (1.5 mrad crossing-angle period, runs 51274–54000). The measurement employs the ABCD double-sideband method for background subtraction, with photon identification based on electromagnetic shower shape variables evaluated by a Boosted Decision Tree (BDT), and topological cluster isolation with $R = 0.4$. Twelve E_{T} bins span 8–35 GeV within $|\eta| < 0.7$.

Since the preliminary result, four areas have undergone substantial development:

1. **BDT photon identification** (§2): Feature expansion, model variant studies, parametric threshold, and systematic framework.

2. **NPB rejection** (§3): New dedicated XGBoost model replacing the hard timing preselection.

3. **Double-interaction efficiency** (§4): Decomposition of the pileup efficiency drop into matching artifact vs. genuine physics effects.

4. **Shower shape comparisons** (§5): Pileup-blended MC templates and quantitative data–MC agreement assessment.

2 BDT Photon Identification

2.1 Feature Expansion

The preliminary BDT used a minimal feature set of 6 variables: cluster E_{T}, η, ϕ, vertex z, $E_{1\times1}/E_{3\times3}$, and $E_{3\times2}/E_{3\times5}$. The updated analysis expands the training feature space to 25 variables, adding tower energy fractions, cluster width moments, and a broader set of energy ratios:

- **Tower energy fractions**: $e_{t1} = (E_1 + E_2 + E_3 + E_4)/E_{\mathrm{tot}}$, e_{t2}, e_{t3}, e_{t4} (quadrant asymmetries)

- **Cluster width moments**: w_η^{CoG} (eta width), w_ϕ^{CoG} (phi width), $w_{3\times2}$, $w_{5\times2}$, $w_{7\times2}$

- **Energy ratios**: $E_{1\times1}/E_{2\times2}$, $E_{1\times1}/E_{1\times3}$, $E_{1\times1}/E_{1\times5}$, $E_{1\times1}/E_{1\times7}$, $E_{1\times1}/E_{3\times1}$, $E_{1\times1}/E_{5\times1}$, $E_{1\times1}/E_{7\times1}$, $E_{2\times2}/E_{3\times3}$, $E_{2\times2}/E_{3\times5}$, $E_{2\times2}/E_{3\times7}$, $E_{2\times2}/E_{5\times3}$

Training uses PYTHIA-8 photon MC (photon5/10/20, truth pid $\in \{1, 2\}$) as signal and jet MC (jet10/15/20/30/50, pid $\notin \{1, 2\}$) as background, with $6 < E_{\mathrm{T}} < 40$ GeV and $|\eta| < 0.7$.

2.2 Model Variants

To study feature importance and optimize the trade-off between discrimination power and robustness, 11 model variants are trained via systematic feature ablation:

Table 1: BDT model variants defined by feature subsets. All variants include vertex z, η, $E_{1\times1}/E_{3\times3}$, and e_{t1}–e_{t4} as baseline features. The "E" suffix denotes variants that include cluster E_{T} as an explicit feature.

Variant	Additional features	Count	Purpose
`base`	(baseline only)	7	Minimal reference
`base_vr`	(vertex-reweighted)	7	Vertex robustness check
`base_v0`	Drops e_{t1}	6	Remove isolation-correlated feature
`base_v1`	$+\,w_\eta^{\mathrm{CoG}}$	8	Add width
`base_v2`	$+\,w_\eta^{\mathrm{CoG}}$, w_ϕ^{CoG}	9	Add 2D shape
`base_v3`	$+\,w_\eta^{\mathrm{CoG}}$, w_ϕ^{CoG}, $E_{3\times2}/E_{3\times5}$	10	Full shape
`base_E`	$+\,E_{\mathrm{T}}$	8	Energy-aware base
`base_v0E`	$+\,E_{\mathrm{T}}$, drops e_{t1}	7	Energy + iso-insensitive
`base_v1E`	$+\,E_{\mathrm{T}}$, w_η^{CoG}	9	**Nominal**
`base_v2E`	$+\,E_{\mathrm{T}}$, w_η^{CoG}, w_ϕ^{CoG}	10	Energy + 2D shape
`base_v3E`	$+\,E_{\mathrm{T}}$, w_η^{CoG}, w_ϕ^{CoG}, $E_{3\times2}/E_{3\times5}$	11	Full feature set

The nominal analysis uses `base_v1E` (9 features) in both E_{T} bins $[8, 15]$ and $[15, 35]$ GeV, with `base_E` as the default fallback.

2.3 Training Procedure

The XGBoost training applies three-stage reweighting to minimize kinematic biases:

1. **Class reweighting**: Balances signal and background counts.
2. **E_{T} spectrum flattening**: Inverse-PDF spline reweighting to remove the steeply falling E_{T} spectrum, preventing the BDT from learning E_{T}-dependent shortcuts.
3. **η spectrum flattening**: Same procedure for the η distribution.

Background subsampling is applied below $E_{\mathrm{T}} = 15$ GeV to address the heavy-tail problem in the low-E_{T} region, using inverse-PDF reweighting across 20 E_{T} bins.

Table 2: XGBoost hyperparameters for the photon identification BDT.

Parameter	Value
Number of boosted trees	750
Maximum depth	5
Learning rate	0.1
Subsample fraction	0.5
Column sample per tree	0.6
L1 regularization (α)	5.0
L2 regularization (λ)	0.3
Grow policy	Loss-guided
Tree method	Histogram-based
Train / validation / test split	70% / 10% / 20%
Random seed	42

A split-sample approach divides MC into two independent halves to avoid self-training bias: each half is scored by the model trained on the other half. Models are exported via `TMVA::Experimental::SaveXGBoost` and applied in the analysis via `TMVA::Experimental::RBDT::Compute()`.

2.4 Training Performance

The nominal `base_v1E` model achieves test-set AUC values of 0.88–0.93 across all E_{T} bins, evaluated on the 20% held-out test sample with cross-section-weighted events. Table 3 summarizes the per-bin performance; the best discrimination is in the 10–15 GeV bin (AUC = 0.925), where the signal-to-background shower-shape contrast is strongest, while the lowest AUC is at $6 < E_{\mathrm{T}} < 10$ GeV (0.884), where fragmentation photons broaden the signal shower profiles.

Table 3: Test-set AUC for the nominal `base_v1E` BDT in each E_{T} training bin. A single global model is trained on all bins simultaneously; the per-bin AUC is evaluated on the test split restricted to each E_{T} range.

E_{T} bin [GeV]	6–10	10–15	15–20	20–25	25–35
Test AUC	0.884	0.925	0.911	0.914	0.912
Test events	591 k	658 k	442 k	389 k	125 k

Figures 1 and 2 show the ROC curves for all 11 photon-ID variants on the `CLUSTERINFO_CEMC` (split) and `CLUSTERINFO_CEMC_NO_SPLIT` (nosplit) cluster nodes respectively, with each panel showing the five E_{T} training bins as separate curves. The curves are well separated from the diagonal chance line across the full false-positive-rate range, confirming that every variant provides strong photon–jet discrimination in the high-E_{T} bins; the $6 < E_{\mathrm{T}} < 10$ GeV bin trails because signal photons at low E_{T} have poorer shower resolution and the jet background is not yet kinematically suppressed. The `base_vr` panel is bit-identical to `base` (consistent with the BDT-training audit's WARN-1: the "vertex reweight" variant config accidentally does not enable `vertex_reweight`, so the trained model is a duplicate of `base`). The `v0` and `v0E` variants, which drop the dominant e_{t1} feature, show the weakest discrimination. AUCs in these plots are evaluated on a ~ 590 k signal + bkg MC subset drawn from `photon5/10/20` and `jet20/30` with $6 < E_{\mathrm{T}} < 40$ GeV and $|\eta| < 0.7$; absolute values therefore differ slightly from the per-bin test-set AUCs quoted in Table 3 (which use the training-time 20% hold-out and class-balance-reweighted events), but the ranking and shape of the curves are preserved.

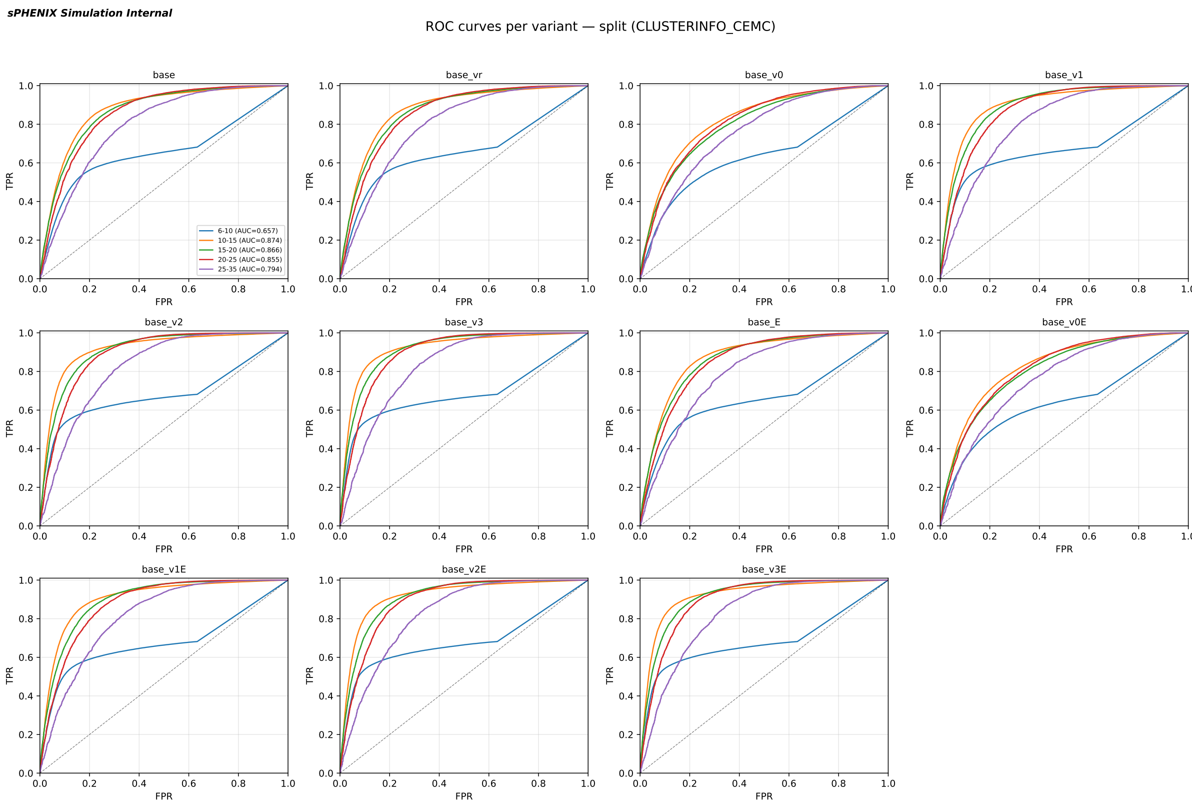


Figure 1: ROC curves for all 11 photon-ID variants on the split cluster node. Each panel shows the five E_{T} training bins (colored curves, legend in the `base` panel). The dashed line indicates chance-level discrimination.

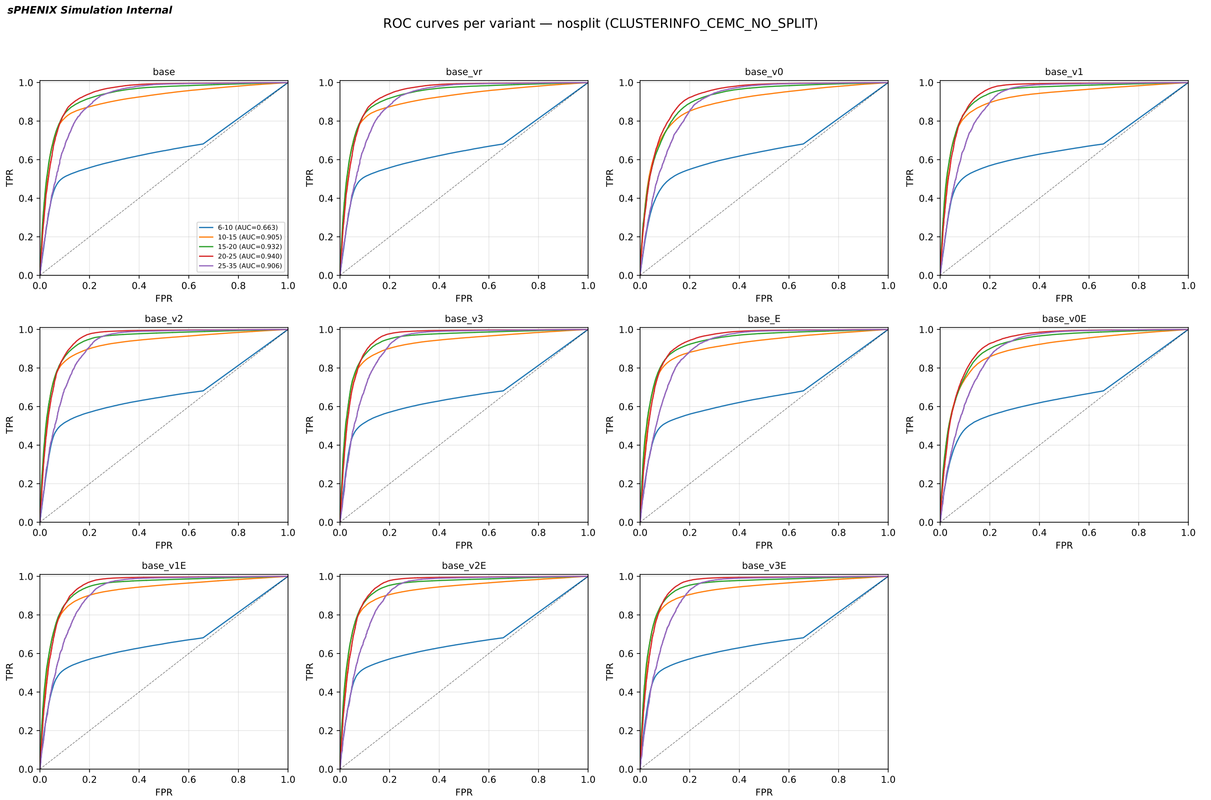


Figure 2: As Figure 1 but for the nosplit cluster node. AUCs are systematically higher than the split case (consistent with the ≈ 5 percentage-point advantage of the nosplit cluster node in Table 3).

The BDT score distributions for signal (photon MC) and background (jet MC) are shown in Figures 3 and 4 for all 11 variants, in the representative $15 < E_{\mathrm{T}} < 20$ GeV bin. Background clusters peak sharply near zero in all variants with the expected shower-shape discrimination; signal photons accumulate at high scores > 0.7. The `base_v0` and `base_v0E` panels show much broader signal distributions extending down to BDT ≈ 0.3, reflecting the weaker discrimination power of dropping e_{t1}. `base_vr` is once again identical to `base`.

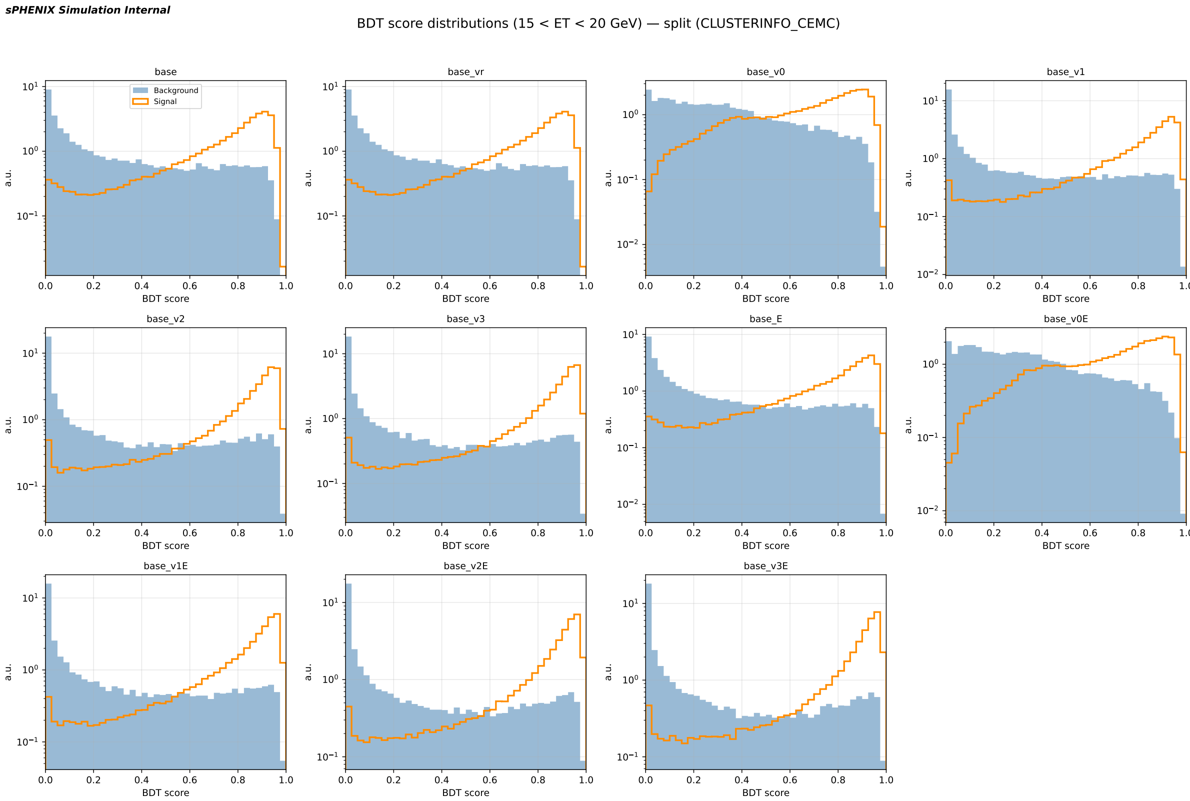


Figure 3: BDT score distributions for signal (orange outline) and background (blue filled) in the $15 < E_{\mathrm{T}} < 20$ GeV bin, for all 11 variants on the split cluster node. Histograms are unit-normalized; log scale on the vertical axis.

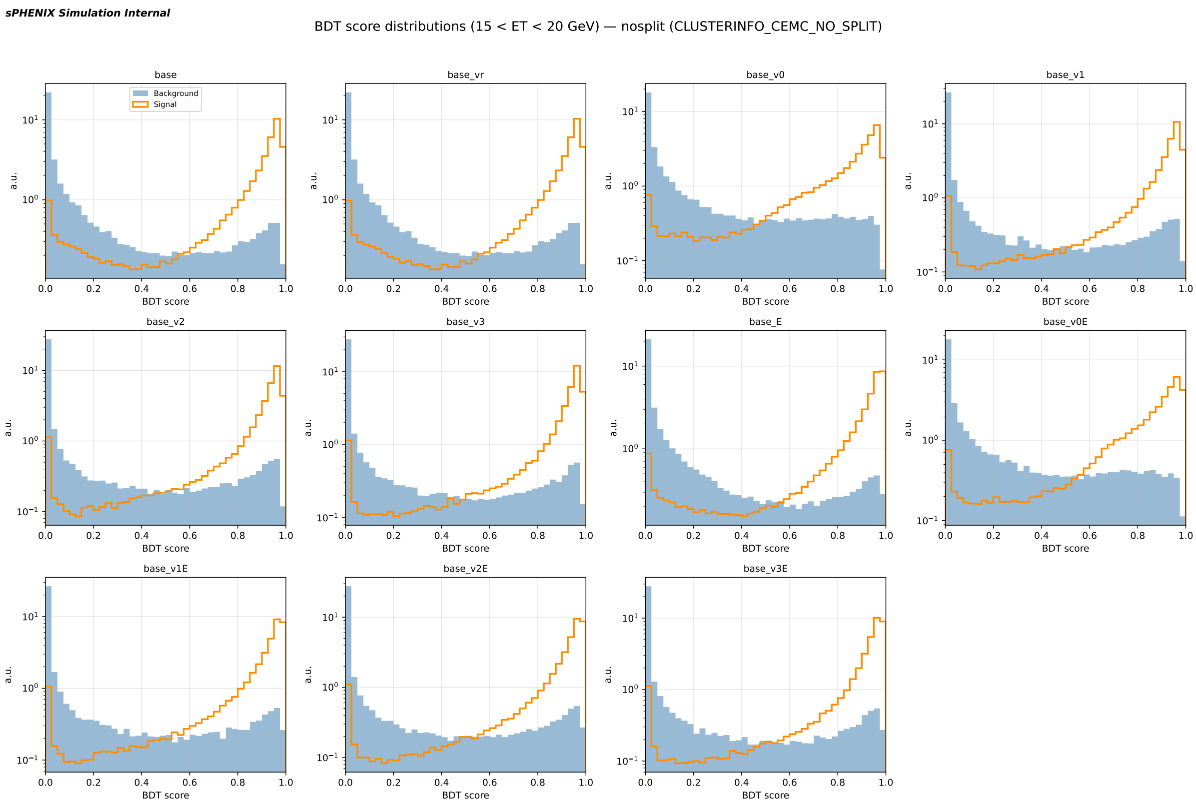


Figure 4: As Figure 3 but for the nosplit cluster node.

Figure 5 shows the XGBoost feature importance (gain metric) for the nine input features of `base_v1E`, broken down by E_T bin. The tower energy fraction e_{t1} dominates the gain, contributing 36% of the total split improvement. The quadrant asymmetries (e_{t2}, e_{t3}), vertex z, and w_η^CoG provide the next tier of discrimination, while $E_{1\times1}/E_{3\times3}$ and cluster η contribute at the 5% level. The stability of the ranking across E_T bins confirms that the model does not rely on E_T-dependent shortcuts.

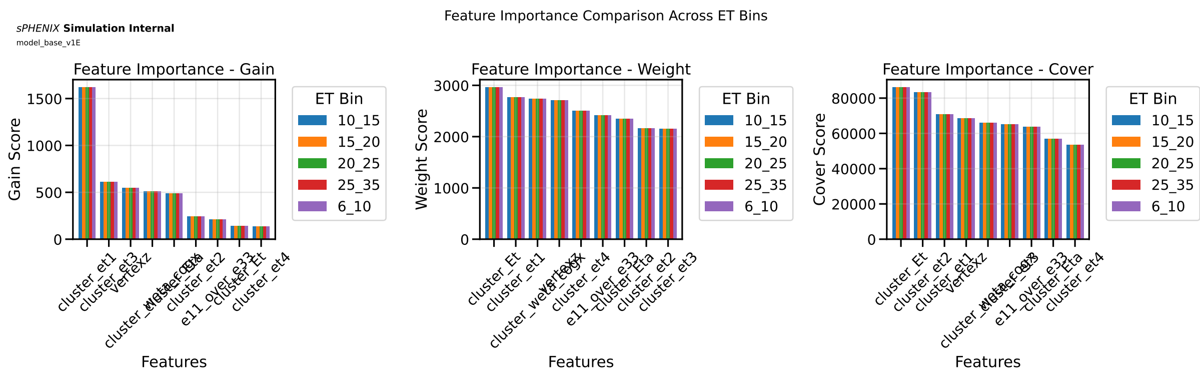


Figure 5: Feature importance for the nominal **base_v1E** model, shown as gain (left), weight (center), and cover (right) across the five E_T training bins. The tower fraction e_{t1} provides the largest gain contribution in all bins.

An important consideration for the ABCD background subtraction is the correlation between the BDT score (defining the tight/non-tight axis) and the isolation energy $E_\mathrm{T}^\mathrm{iso}$ (defining the isolated/non-isolated axis). Figure 6 shows the 2D scatter of $E_\mathrm{T}^\mathrm{iso}$ vs. BDT score for signal and background clusters. The overall Spearman correlation coefficient is $\rho = -0.585$, driven primarily by the background population where non-isolated jets naturally receive low BDT scores. The signal photons cluster in the upper-left corner (high BDT, low E_T^{iso}), well separated from the background. The residual correlation is accounted for in the ABCD method through the signal leakage corrections c_B, c_C, and c_D.

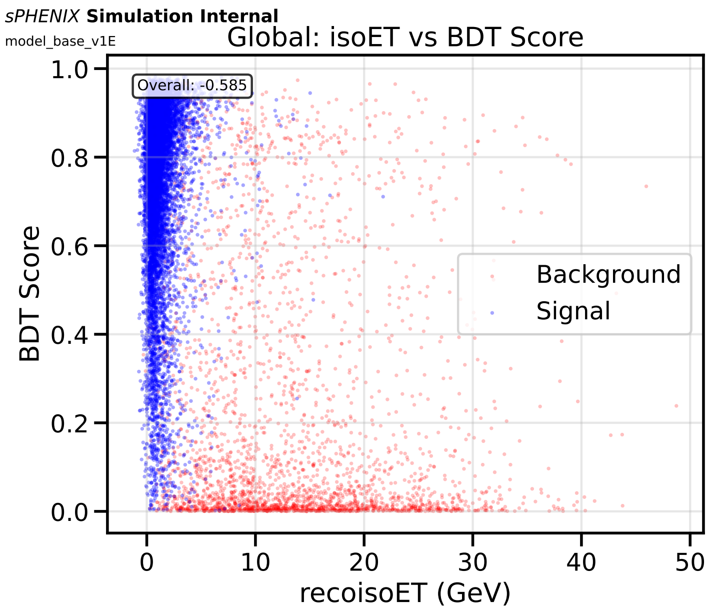


Figure 6: Scatter plot of reconstruction-level isolation energy E_T^{iso} vs. BDT score for signal (blue) and background (red) clusters, nominal `base_v1E` model. The overall Spearman correlation is $\rho = -0.585$. Signal photons are concentrated at high BDT score and low E_T^{iso}, while background jets populate the low-score, high-E_T^{iso} region.

To make the iso vs. BDT behaviour easier to read across models, the same distribution is re-plotted with the axes swapped (BDT score on the horizontal axis, E_T^{iso} on the vertical) and a binned profile of the mean E_T^{iso} per BDT-score bin overlaid for signal and background separately. Figures 7 and 8 show this profile view for all 11 photon-ID variants on the `CLUSTERINFO_CEMC` (split) and `CLUSTERINFO_CEMC_NO_SPLIT` (nosplit) cluster nodes respectively. In every panel the signal profile drops from ≈ 5 GeV at the low-BDT tail to $\lesssim 2$ GeV above BDT ≈ 0.6, while the background profile stays near ~ 10 GeV with only a shallow slope — confirming that the BDT defines an essentially ABCD-usable axis for all variants, with the residual correlation absorbed into the signal-leakage corrections c_B, c_C, c_D. The `base_vr` panel is bit-identical to `base` across both cluster nodes, re-confirming that the "vertex reweight" variant is currently dead (the `reweighting.vertex_reweight: true` override is never injected into the generated variant configs — see the BDT training audit note for details).

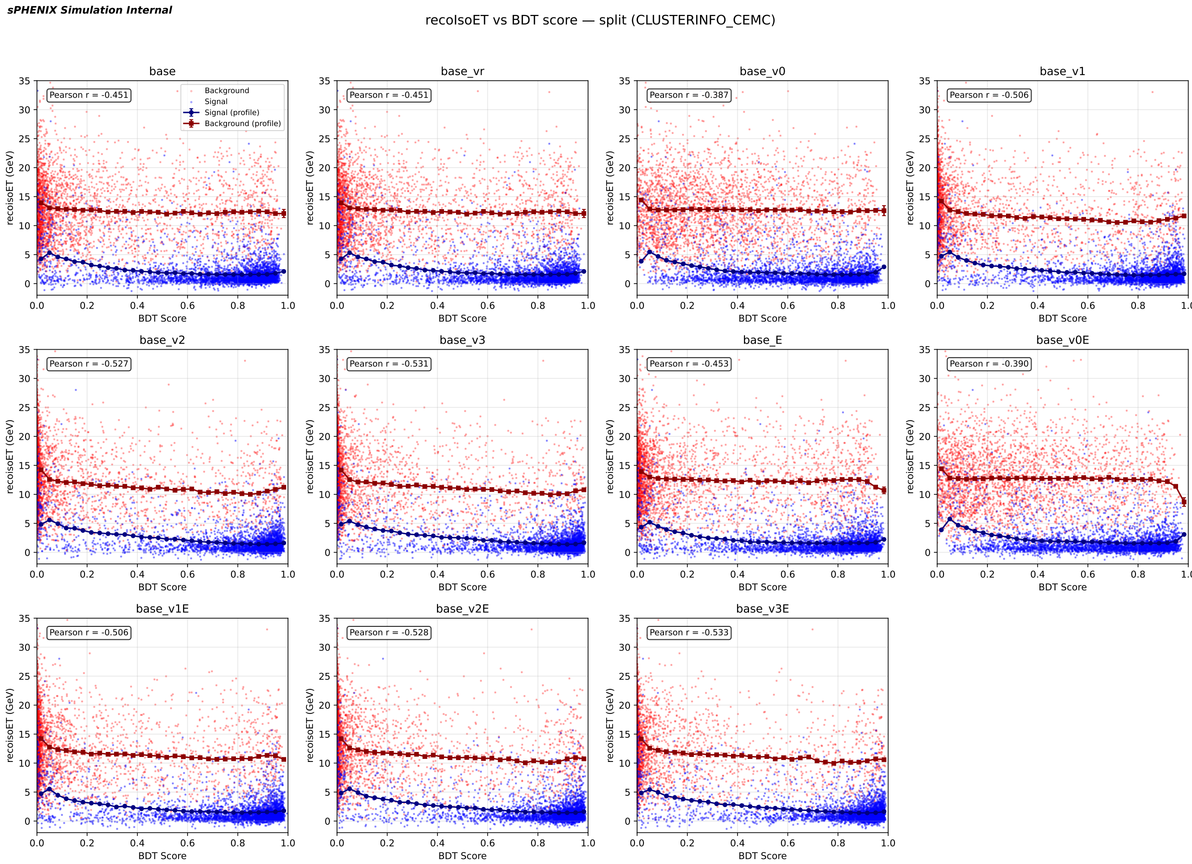


Figure 7: $E_{\mathrm{T}}^{\mathrm{iso}}$ vs. BDT score profile for all 11 photon-ID variants on the split cluster node (`CLUSTERINFO_CEMC`). In each panel, 4000 random clusters per class are scatter-plotted (signal blue, background red); connected markers show the binned profile mean $\pm$ SEM of $E_{\mathrm{T}}^{\mathrm{iso}}$ per BDT-score bin (navy: signal, dark red: background). The overall Pearson r (combined pool) is quoted in each panel. Statistics: $\sim 380\,\mathrm{k}$ signal and $\sim 210\,\mathrm{k}$ background clusters from `photon5/10/20` and `jet20/30` MC after $6 < E_{\mathrm{T}} < 40$ GeV, $|\eta| < 0.7$.

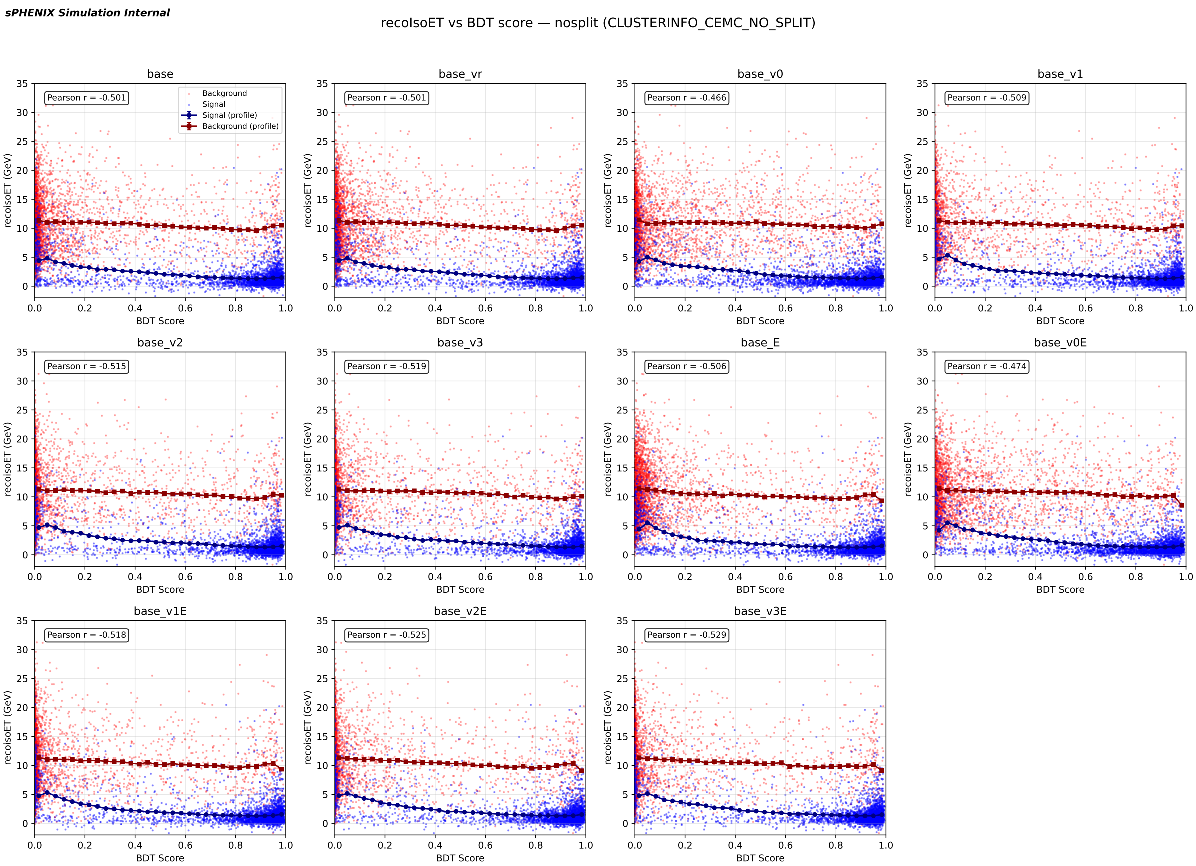


Figure 8: As Figure 7 but for the nosplit cluster node (`CLUSTERINFO_CEMC_NO_SPLIT`). The signal profile drops more steeply with BDT score and the background profile sits slightly lower than in the split case, consistent with the ~ 5 percentage-point higher test AUCs of the nosplit models.

2.5 E_T-Dependent Parametric Threshold

The preliminary analysis used a flat BDT threshold. The updated analysis implements an E_T-dependent parametric cut:

$$\mathrm{BDT}_{\mathrm{min}}^{\mathrm{tight}} = 0.80 - 0.015 \times E_\mathrm{T}\,[\,\mathrm{GeV}], \tag{1}$$

where the intercept and slope are configurable in the analysis YAML. This accommodates the improving signal-to-background ratio at higher E_T, allowing a looser BDT requirement where the background contamination is naturally lower.

Clusters with $E_\mathrm{T} < 7$ GeV receive a default BDT score of -1 and are excluded from the analysis.

2.6 BDT Systematic Variations

The systematic uncertainty framework for the BDT is implemented through automated config generation (`make_bdt_variations.py`), producing 30+ analysis configurations. Key BDT-related variations:

Table 4: BDT-related systematic variations. Each variation generates a separate analysis config with a unique `var_type` suffix.

Variation	Parameter change	Syst. type	Role
`tightbdt50`	Intercept → 0.70	Efficiency	Down
`tightbdt70`	Intercept → 0.90	Efficiency	Up
`ntbdtmin05`	Non-tight lower → 0.05	Purity	One-sided
`ntbdtmin10`	Non-tight lower → 0.10	Purity	One-sided
`etbin_v3E_v3E`	Both bins use `base_v3E`	Efficiency	Max-envelope
`etbin_E_E`	Both bins use `base_E`	Efficiency	Max-envelope
`etbin_v1E_v3E`	Mixed models	Efficiency	Max-envelope

The tight BDT threshold variation yields 5–10% relative systematic uncertainty on the cross-section. The non-tight lower boundary variation contributes up to 15% relative uncertainty, primarily affecting the purity estimate through the ABCD sideband definition.

3 Non-Physics Background Rejection

3.1 From Hard Cut to BDT Score

The preliminary analysis rejected non-physics background (NPB) clusters using a hard timing preselection: clusters with $t_{\text{cluster}} - t_{\text{MBD}} < -5$ ns were tagged as NPB and removed. This approach has two limitations: (1) the timing threshold is somewhat arbitrary, and (2) it provides no discrimination for NPB clusters that happen to arrive near $t = 0$.

The updated analysis replaces this with a dedicated 25-feature XGBoost model that scores each cluster on a continuous $[0, 1]$ scale, where high scores indicate physics clusters and low scores indicate NPB.

3.2 NPB Score Training

The NPB BDT is trained on:

- **Signal (physics)**: Clusters from PYTHIA jet MC (jet10, jet15, jet20, jet30, jet50) with valid particle ID, $6 < E_{\text{T}} < 40$ GeV, $|\eta| < 0.7$.
- **Background (NPB)**: Clusters from collision data with $t_{\text{cluster}} - t_{\text{MBD}} < -7$ ns, which are overwhelmingly beam-induced backgrounds not synchronized with the collision time.

The model uses 25 input features: cluster E_{T}, η, vertex z, ten tower energy ratios ($E_{1\times1}/E_{3\times3}$, $E_{3\times2}/E_{3\times5}$, $E_{1\times1}/E_{2\times2}$, $E_{1\times1}/E_{1\times3}$, etc.), cluster width moments (w_η^{CoG}, w_ϕ^{CoG}, $w_{3\times2}$, $w_{5\times2}$, $w_{7\times2}$), and tower energy fractions (e_{t1}–e_{t4}).

Table 5: XGBoost hyperparameters for the NPB score model. Note the differences from the photon ID BDT (Table 2): deeper trees and less aggressive regularization, reflecting the cleaner signal–background separation for NPB rejection.

Parameter	Value
Number of boosted trees	750
Maximum depth	6
Learning rate	0.1
Subsample fraction	0.8
Column sample per tree	0.8
L1 regularization (α)	1.0
L2 regularization (λ)	1.0

The trained model is exported to ROOT TMVA format and applied in `apply_BDT.C`, producing a new branch `cluster_npb_score_CLUSTERINFO_CEMC` for each cluster with $6 < E_T < 40$ GeV and $|\eta| < 0.7$.

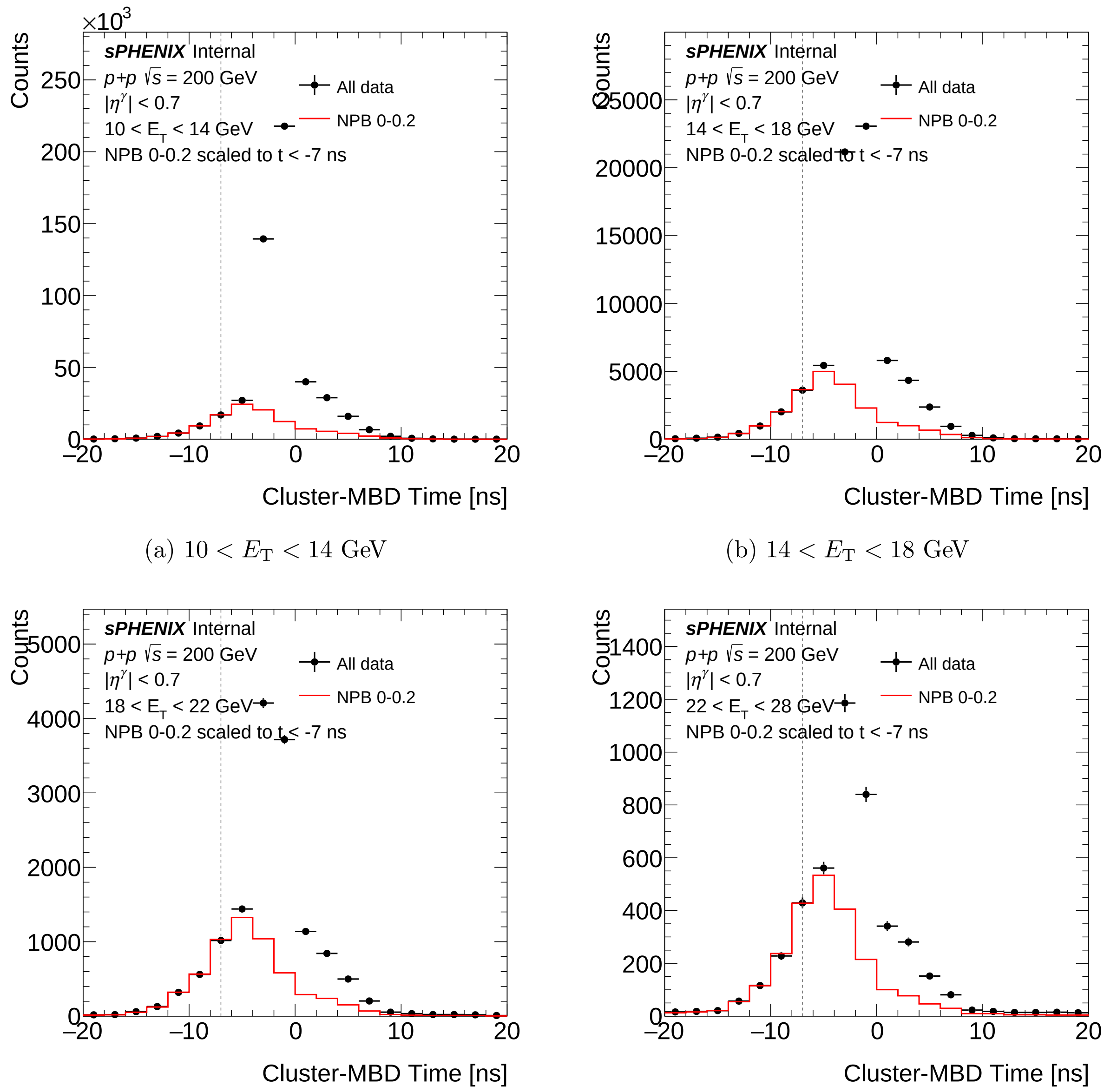
3.3 NPB Score Validation

The NPB score is validated independently using the cluster–MBD timing distribution $t_{\text{cluster}} - t_{\text{MBD}}$:

- At NPB score > 0.5: the timing distribution is sharply peaked near $t = 0$ ns, confirming physics origin.
- At NPB score < 0.2: the distribution is broad, extending to large negative times ($t < -7$ ns), confirming NPB origin.

A background template is constructed from the out-of-time tail ($t < -7$ ns) and normalized to estimate the NPB fraction as a function of the score threshold.

Figure 9 shows the cluster–MBD time distributions in data for three representative E_T bins, with the low-NPB-score background template overlaid. The template (normalized in the $t < -7$ ns tail) matches the out-of-time component, demonstrating that the NPB score correctly identifies beam-background clusters independently of the timing variable used to define the training labels.



(a) $10 < E_{\mathrm{T}} < 14$ GeV

(b) $14 < E_{\mathrm{T}} < 18$ GeV

(c) $18 < E_{\mathrm{T}} < 22$ GeV

(d) $22 < E_{\mathrm{T}} < 28$ GeV

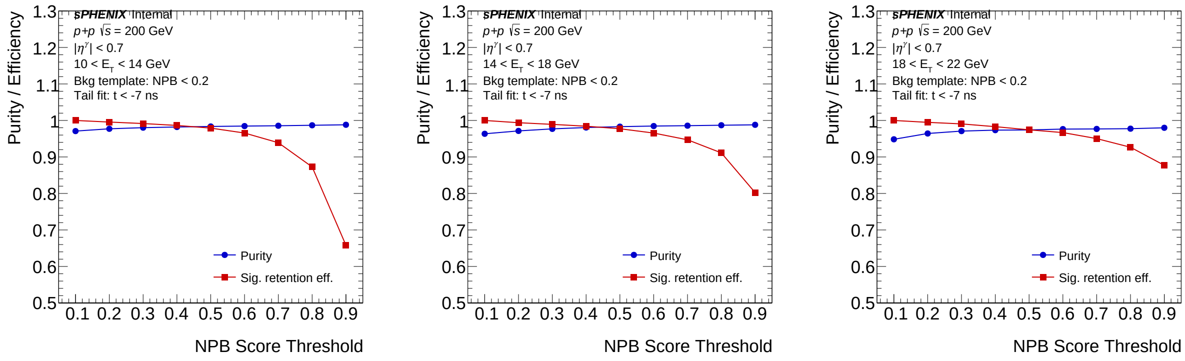
Figure 9: Cluster–MBD time distributions in data (black) with low-NPB-score background template (red, normalized in $t < -7$ ns tail) for four E_{T} bins. The background template tracks the out-of-time tail, confirming the NPB score identifies beam backgrounds independent of timing.

3.4 Cut Optimization

The nominal threshold is set at NPB score > 0.5, achieving:

- **NPB purity**: $\sim$99% of surviving clusters are physics (NPB contamination $\sim$1–2%).
- **Signal retention**: $\sim$98% of MC physics clusters pass the threshold.

Figure 10 shows the NPB purity (fraction of physics clusters) and cluster yield as a function of the NPB score threshold for three E_{T} bins. The purity rises steeply from $\sim$96% at threshold 0.1 to $\sim$99% at the nominal 0.5, with minimal yield loss. Figure 11 shows the corresponding MC selection efficiency.



(a) 10–14 GeV (b) 14–18 GeV (c) 18–22 GeV

Figure 10: Physics-cluster purity as a function of NPB score threshold for three E_{T} bins. Purity is estimated via timing-based background subtraction ($t < -7$ ns template). The nominal threshold at 0.5 achieves ∼99% purity across all bins.

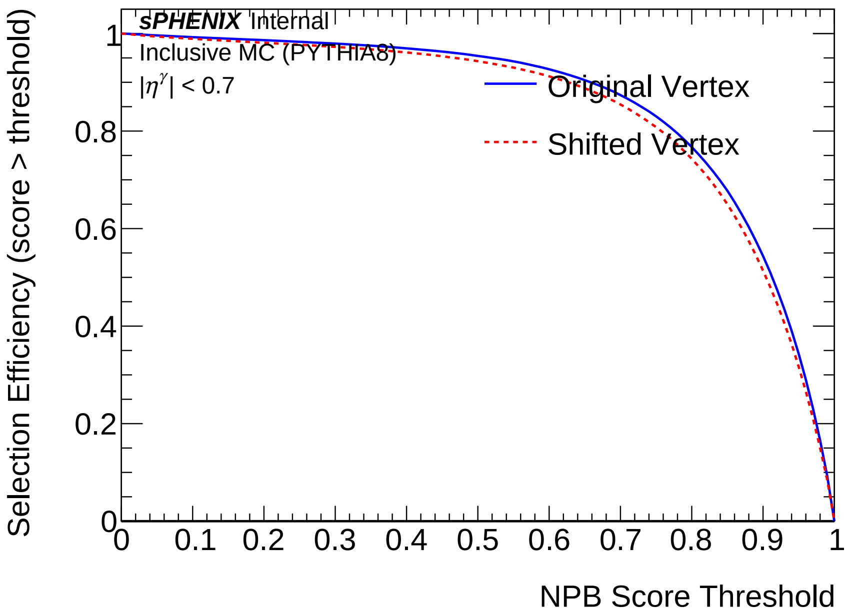


Figure 11: Selection efficiency of the NPB score cut on inclusive MC clusters as a function of the NPB threshold. At the nominal threshold of 0.5, ∼98% of MC physics clusters are retained.

3.5 NPB Systematic Variations

The NPB score threshold is varied as a systematic:

- `npb03`: Threshold → 0.3 (looser, higher NPB contamination)
- `npb07`: Threshold → 0.7 (tighter, lower signal retention)

The cross-section deviations from the nominal are taken as bin-by-bin systematic uncertainties.

4 Double-Interaction Efficiency Study

4.1 Motivation

When double-interaction MC samples (produced with full GEANT4 simulation of two overlapping $p+p$ collisions) were passed through the original efficiency pipeline (which required $\Delta R < 0.1$ geometric truth-matching on top of the GEANT track ID), the reconstruction efficiency dropped dramatically: from ∼91% (single interaction) to ∼31% (double interaction). A dedicated investigation demonstrated that this drop was overwhelmingly a truth–reco matching artifact caused by vertex displacement. Based on this finding, the $\Delta R < 0.1$ requirement was removed from the production efficiency calculation; track ID matching alone provides the definitive truth association. This section documents both the artifact diagnosis and the corrected efficiency breakdown after ΔR removal.

4.2 Pileup Fraction Calculation

The mean number of collisions per bunch crossing is computed from the Poisson-corrected MBD coincidence rate:

$$\mu = \frac{\sum_{\text{runs}} N_{\text{MBD,corr}}}{r_B \cdot \sum_{\text{runs}} \Delta t}, \tag{2}$$

where $r_B = 111 \times 78{,}200 = 8.68 \times 10^6$ Hz is the RHIC bunch crossing rate. The event-level interaction fractions are:

$$f_1 = \frac{\mu\, e^{-\mu}}{1 - e^{-\mu}}, \qquad f_2 = \frac{(\mu^2/2)\, e^{-\mu}}{1 - e^{-\mu}}, \qquad f_{\geq 3} = 1 - f_1 - f_2. \tag{3}$$

Fractions are computed per run and luminosity-weighted (not simple-averaged) to avoid Jensen's inequality bias. Cluster-weighted fractions account for the higher cluster multiplicity in multi-interaction events:

$$w_{\text{single}} = \frac{f_1}{\sum_k k\, f_k}, \qquad w_{\text{double}} = \frac{2\, f_2 + 3\, f_{\geq 3}}{\sum_k k\, f_k}. \tag{4}$$

Table 6: Luminosity-weighted pileup fractions and cluster-level mixing weights.

Crossing angle	Run range	μ_{corr}	f_1	f_2	w_{single}	w_{double}
0 mrad	47289–51274	0.24	0.879	0.111	0.776	0.224
1.5 mrad	51274–54000	0.08	0.960	0.039	0.921	0.079

4.3 Truth Photon Content

A prerequisite for interpreting the efficiency drop is verifying that the double-interaction MC does not inject additional truth-level prompt photons from the second collision.

Table 7: Truth-level prompt photon content per event (photon10 samples, 100k events each). The denominator requires $|\eta^\gamma| < 0.7$ and $E_{\text{T}}^{\text{iso,truth}} < 4$ GeV.

Quantity	Single MC	Double MC
Mean truth photons per event	0.493	0.495
Events with ≥ 2 denom. photons	0.22%	0.21%
Mean truth $E_{\text{T}}^{\text{iso,truth}}$ [GeV]	0.17	0.26

The truth photon content is statistically identical. The slightly higher mean truth isolation energy in double MC (0.26 vs. 0.17 GeV) reflects additional hadrons from the second collision falling within the isolation cone, but this is well below the 4 GeV fiducial threshold.

4.4 Vertex Shift Mechanism

In double-interaction events, the reconstructed vertex is the average of the two collision points:

$$z_{\mathrm{vtx}}^{\mathrm{reco}} = \frac{z_{\mathrm{vtx},1} + z_{\mathrm{vtx},2}}{2}. \tag{5}$$

Since both vertices are drawn from a broad distribution (RMS ∼30 cm), the displacement $\Delta z = z_{\mathrm{vtx}}^{\mathrm{reco}} - z_{\mathrm{vtx}}^{\mathrm{truth}}$ can be large. The cluster pseudorapidity shift at EMCal radius $R_{\mathrm{EMCal}} = 93.5$ cm is:

$$\Delta\eta \approx -\frac{\Delta z}{R_{\mathrm{EMCal}} \cosh\eta}, \tag{6}$$

which at $\eta = 0$ gives $|\Delta\eta| \approx |\Delta z|/93.5$ cm. The $\Delta R < 0.1$ matching threshold therefore fails when $|\Delta z| \gtrsim 9.35$ cm at midrapidity.

Table 8: Vertex shift distributions. The critical threshold $|\Delta z| = 9.35$ cm corresponds to $|\Delta\eta| = 0.1$ at $\eta = 0$.

Quantity	Single MC	Double MC
RMS of Δz [cm]	7.6	34.5
Fraction with $\lvert\Delta z\rvert > 9.35$ cm	—	83%

The sharp transition in matching failure rate as a function of $|\Delta z|$ confirms the deterministic geometric origin:

Table 9: Matching failure rate vs. vertex displacement in double MC.

$\lvert\Delta z\rvert$ range [cm]	Matching failure rate
< 10	9%
10–15	77%
15–20	∼100%
> 20	100%

4.5 Decomposition of the ΔR Matching Failure

For truth photons that failed the $\Delta R < 0.1$ match in double MC, the failure mode was classified by searching for the `trkID`-matched cluster regardless of ΔR:

- **ΔR failure (88%)**: A cluster with the correct `trkID` exists but $\Delta R > 0.1$ due to vertex shift. The signal photon cluster is present — it merely appears at the wrong η when reconstructed from the displaced vertex.
- **Missing cluster (12%)**: No cluster with the matching `trkID` is found, corresponding to genuine cluster loss from merging in the higher-multiplicity environment.
- **Extra truth photons (∼0%)**: The second collision does not produce additional prompt photons.

This decomposition motivated removing the $\Delta R < 0.1$ requirement from the production efficiency calculation.

4.6 Removal of the ΔR Matching Requirement

A dedicated study of single-interaction MC demonstrated that even without pileup, the $\Delta R < 0.1$ cut rejects $\sim$3.5% of correctly `trkID`-matched truth photons (Table 10). The rejection is driven entirely by vertex mismatch ($|\Delta z| > 9.35$ cm), not cluster quality: 73% of rejected clusters have $E_T/p_T^{\text{truth}} \in [0.8, 1.2]$. Six physics arguments support the removal (detailed in the companion report): the track ID is ground truth; the cut tests vertex resolution, not calorimeter performance; the response matrix captures vertex-induced kinematic shifts; and the $\sim$3% bias is vertex-dependent.

Table 10: $\Delta R < 0.1$ failure rates in single-interaction MC. "Fail ΔR" is the fraction of `trkID`-matched photons rejected by the geometric cut.

Sample	Truth photons	No trkID match	Fail $\Delta R < 0.1$ (of matched)
photon5	273,017	3.42%	2.70%
photon10	2,380,500	2.37%	2.98%
photon20	2,840,703	1.72%	4.01%
Combined	5,494,220	2.09%	3.50%

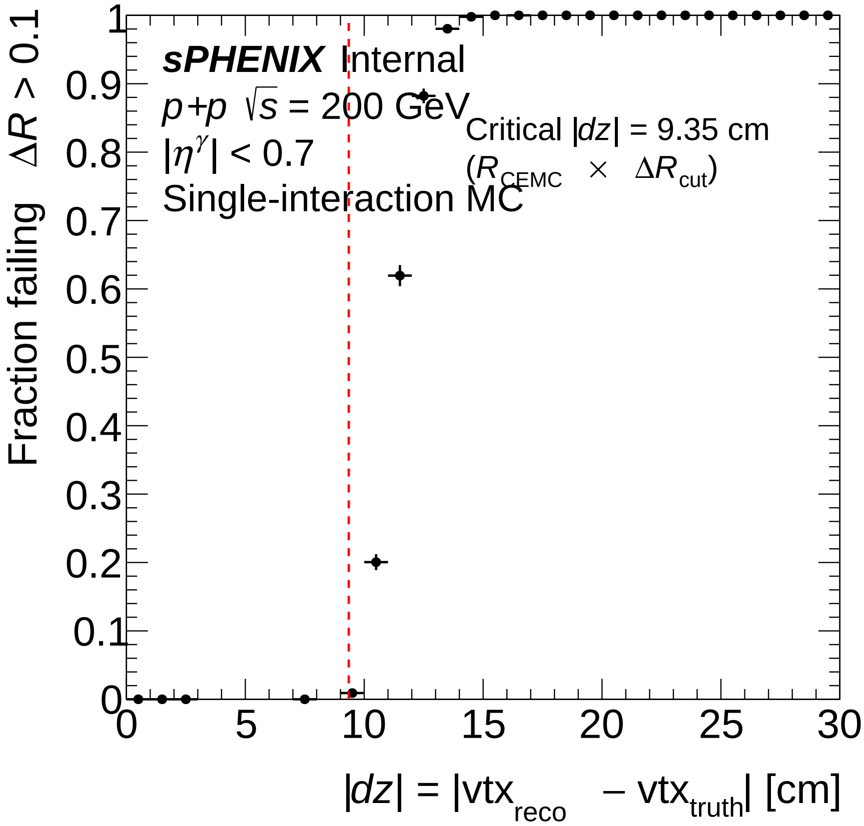


Figure 12: $\Delta R < 0.1$ failure fraction as a function of $|\Delta z|$ in single-interaction MC. The sharp step at $\sim$9.35 cm matches the geometric prediction from the EMCal barrel radius, confirming vertex mismatch as the sole driver of the matching failure.

4.7 Per-Component Efficiency Breakdown (without ΔR)

With the ΔR requirement removed, the factorized efficiency chain $\varepsilon_{\text{reco}} \to \varepsilon_{\text{iso|reco}} \to \varepsilon_{\text{id|reco+iso}} \to \varepsilon_{\text{all}}$ reveals the *genuine* pileup effects at each stage. Table 11 presents the integrated efficiency for single, double, and mixed (0 mrad) MC with `trkID`-only matching.

Table 11: Per-component efficiency with `trkID`-only matching (no ΔR requirement), for photon10 samples integrated over $E_{\text{T}}^{\gamma} = 14$–$30$ GeV. Mixed = 77.6% single + 22.4% double (0 mrad cluster-weighted fractions).

Efficiency component	Single MC	Double MC	Mixed 0 mrad	Effect (single $\to$ double)
$\varepsilon_{\text{reco}}$ (reconstruction)	93.5%	73.1%	88.4%	-20 pp (cluster loss + shifted kinematics
ε_{iso} (isolation | reco)	74.2%	71.0%	73.5%	-3 pp (extra cone energy)
ε_{id} (BDT | reco+iso)	82.7%	69.4%	80.0%	**$-$13 pp** (vertex-shifted kinematics)
ε_{all} (total)	57.4%	36.0%	52.0%	-21 pp (reco + BDT dominate)

Three observations emerge:

- **Reconstruction** ($\varepsilon_{\text{reco}}$): The 20 pp gap between single (93.5%) and double (73.1%) MC reflects genuine cluster loss (no-`trkID`-match rate: 7.9% for double vs. 2.4% for single, accounting for $\sim$5.5 pp) and common-cut failures from shifted kinematics ($\sim$15 pp).
- **Isolation** (ε_{iso}): Essentially unchanged (3 pp degradation). The additional energy from the second collision causes a small increase in isolation cone energy, independent of the ΔR matching.
- **Photon ID** (ε_{id}): **Significant 13 pp degradation** in double MC (69.4% vs. 82.7%). The mechanism is the E_{T}-dependent BDT threshold: vertex displacement shifts the reconstructed E_{T} and η, changing both the BDT working point and the input shower-shape variables (particularly w_{η}^{CoG}, which is sensitive to η). In single MC, the effect is negligible (-0.2 pp) because only $\sim$2.3% of clusters have large vertex shifts. In double MC, $\sim$58% of recovered clusters are "new" (had $\Delta R > 0.1$), and these have a BDT pass rate of only 59.8% (vs. 82.7% for all clusters).

Figure 13 shows the per-stage efficiency as a function of truth p_{T} for all three sample types, providing a clean decomposition of the genuine pileup degradation at each selection stage. Figure 14 shows the BDT pass rate specifically for the "new" clusters recovered by removing the ΔR cut, confirming that vertex-shifted kinematics degrade the BDT score.

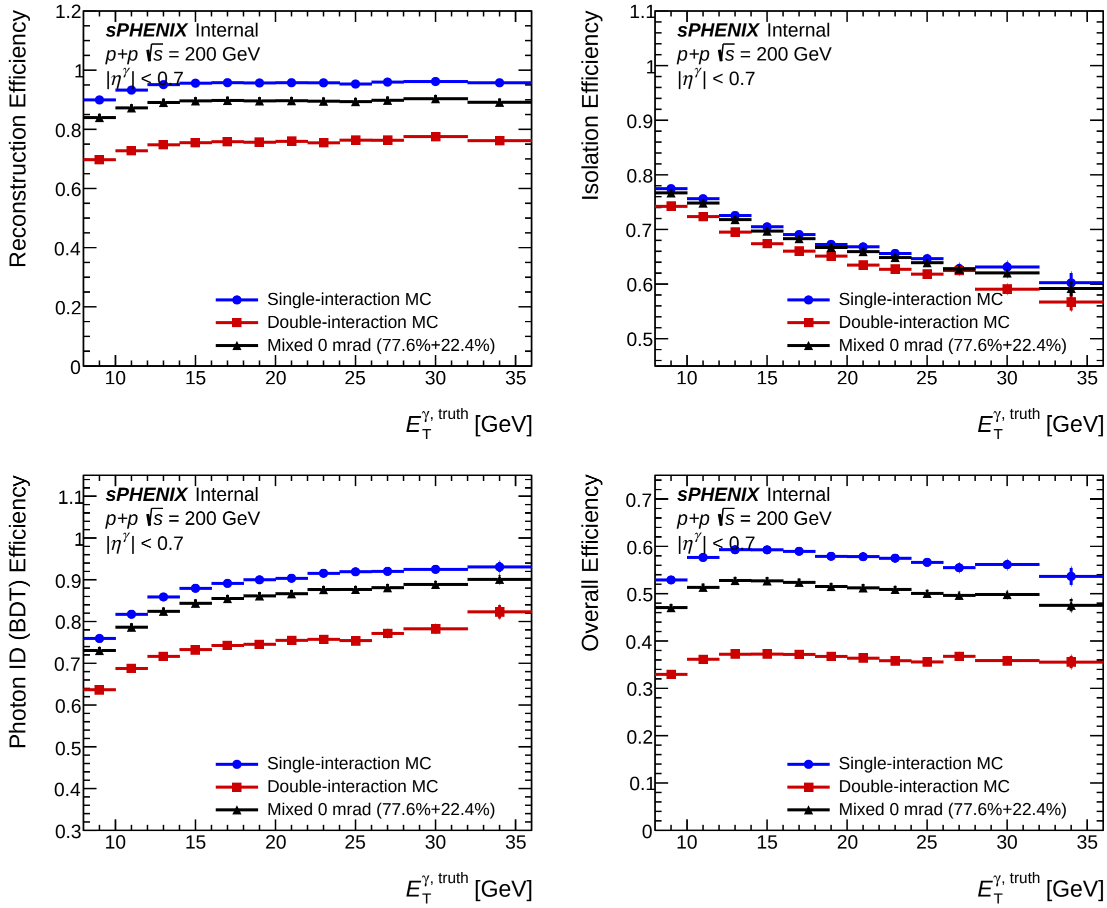


Figure 13: Per-stage efficiency vs. truth p_{T} without the ΔR cut, overlaying single (blue), double (red), and mixed 0 mrad (black) MC. Upper left: reconstruction. Upper right: isolation. Lower left: photon ID (BDT). Lower right: total. The BDT stage shows a clear 13 pp gap between single and double MC, driven by vertex-shifted kinematics.

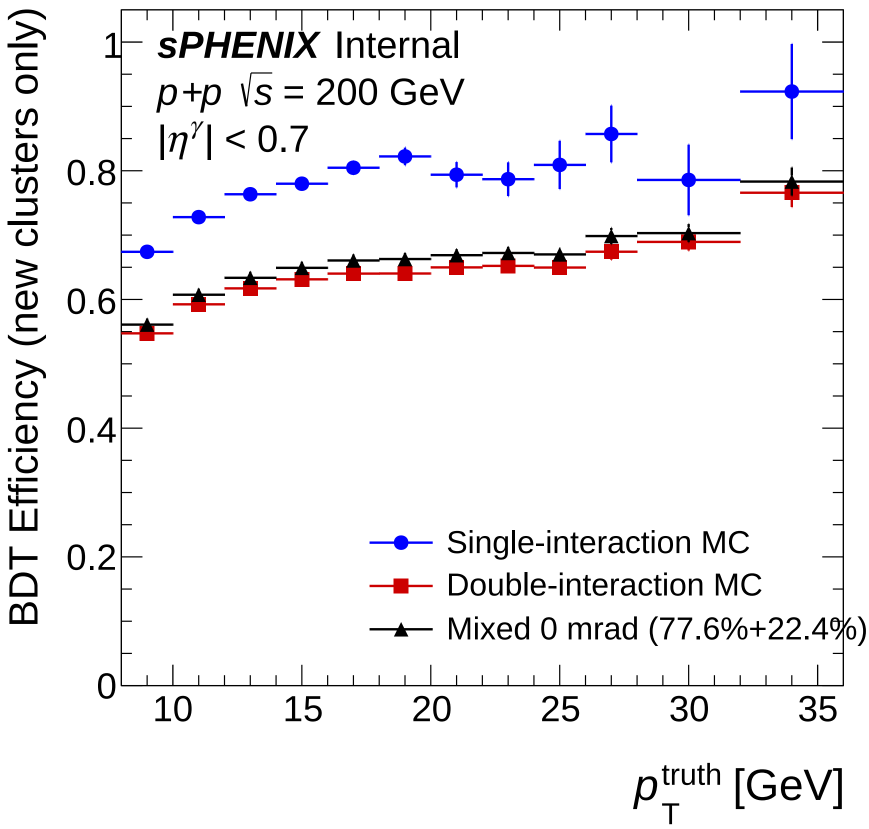


Figure 14: BDT pass rate for "new" clusters (those with $\Delta R > 0.1$ but valid `trkID`) vs. truth p_T. These vertex-shifted clusters have 10–13 pp lower BDT pass rates than the full population, with the deficit narrowing at higher p_T where the BDT score is further above threshold.

For comparison, Table 12 shows the legacy efficiency with the $\Delta R < 0.1$ requirement, which masked the BDT degradation by pre-filtering vertex-shifted clusters.

Table 12: Per-component efficiency with legacy $\Delta R < 0.1$ matching, for reference. The large reco drop is the matching artifact; the BDT appears "unaffected" only because vertex-shifted clusters were already excluded.

Efficiency component	Single MC	Double MC	Mixed 0 mrad
$\varepsilon_{\mathrm{reco}}$ ($\Delta R < 0.1$)	91.4%	30.7%	76.2%
$\varepsilon_{\mathrm{iso}}$	74.2%	71.1%	73.9%
$\varepsilon_{\mathrm{id}}$ (BDT)	82.9%	82.6%	82.9%
$\varepsilon_{\mathrm{all}}$	56.2%	18.0%	46.6%

4.8 `TEfficiency` Weighted-Event Fix

During this investigation, a bug was identified in the `TEfficiency` usage within `RecoEffCalculator_TTreeReade`
The original `SetWeight(w) + Fill(pass, pT)` pattern stores unweighted event counts, biasing the merged efficiency toward the sample with the most raw events. The corrected implementation uses `SetUseWeightedEvents()` with `FillWeighted(pass, weight, pT)`, which stores weighted sums so that `hadd` correctly combines samples by physics weight.

4.9 Implications

With the ΔR requirement removed, the efficiency comparison between single and double MC isolates three genuine pileup effects. For the nominal 1.5 mrad period with 7.9% cluster-weighted pileup:

- **Reconstruction loss**: 20 pp degradation in double MC (93.5% → 73.1%), from genuine cluster loss (∼5.5 pp) and common-cut failures from shifted kinematics (∼15 pp). Weighted by 7.9% pileup fraction: ∼1.6% overall.
- **BDT degradation**: 13 pp in double MC (82.7% → 69.4%), from vertex-shifted E_T and shower-shape inputs to the E_T-dependent BDT threshold. Weighted: ∼1.0% overall.
- **Isolation degradation**: 3 pp in double MC (74.2% → 71.0%), from additional hadronic energy in the isolation cone. Weighted: ∼0.2% overall.
- **Total pileup effect on efficiency**: Mixed 0 mrad efficiency is 52.0% vs. 57.4% for single MC (−9.4% relative). For the 1.5 mrad period, the effect is smaller due to the lower pileup fraction.

The efficiency correction uses single-interaction MC only. The genuine pileup effects are small compared to other systematic uncertainties and are partially absorbed by the mixed MC shower-shape cross-check (§5). Double-interaction MC is reserved for shower-shape template comparisons.

5 Shower Shape Data–MC Comparisons with Pileup Blending

5.1 Two-Pass Blending Pipeline

The pileup mixing uses a two-pass procedure with `ShowerShapeCheck.C`:

Pass 1 (vertex scan): The z_{vtx} distribution is filled for each MC sample weighted by the cluster-level fraction (w_{single} or w_{double} from Table 6). The resulting histograms are merged via `hadd` to produce a blended MC vertex distribution.

Pass 2 (full analysis): All MC samples are reprocessed using the blended vertex distribution from Pass 1 for vertex reweighting, ensuring consistent z_{vtx} templates. Each sample's fills are scaled by its mixing weight multiplicatively on top of the cross-section weight. Single and double outputs are merged to produce the final blended MC.

The MC samples are photon10 (signal, 14–30 GeV truth p_T) and jet12 (background, 14–21 GeV), each with a corresponding `_double` sample.

5.2 Comparison Methodology

Shower shape distributions are compared at two selection levels:

- **cut0**: No selection. Data includes NPB contamination not modeled in MC, complicating interpretation.
- **cut1**: After NPB preselection (NPB score > 0.5). Removes non-physics clusters, providing a cleaner pileup assessment. **This is the preferred comparison level.**

Each comparison plot shows data (black), signal MC (red), and inclusive MC (blue), area-normalized, with χ^2/ndf and a Data/Inclusive MC ratio panel.

5.3 Results: 0 mrad Period

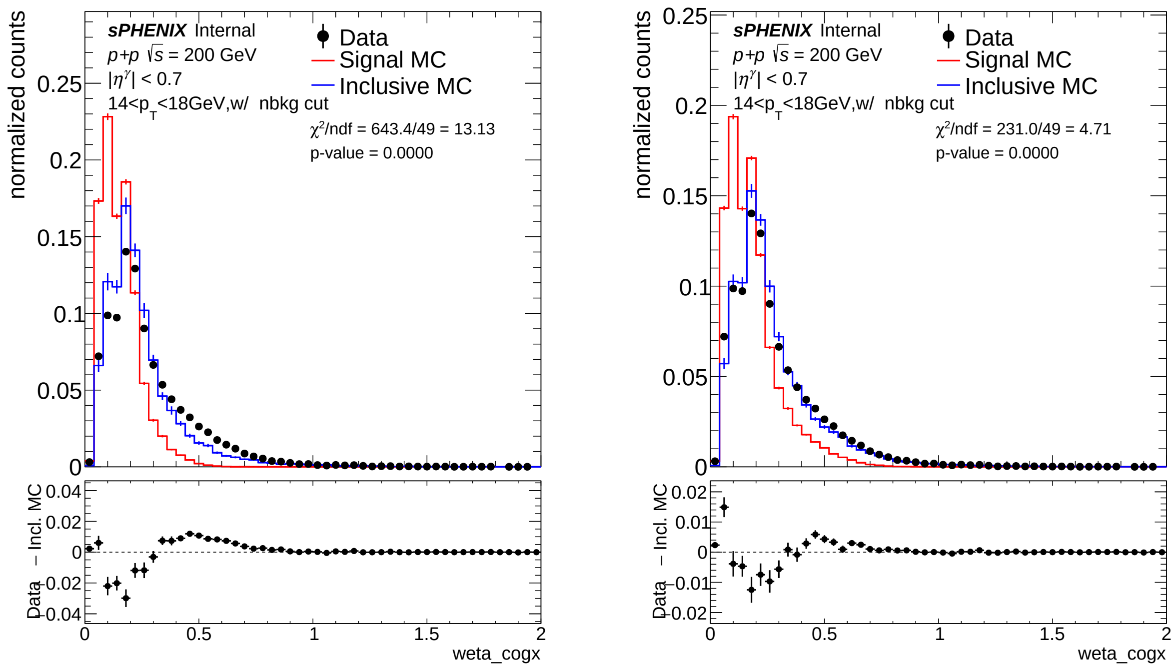
Table 13 compiles the χ^2/ndf values at cut1 for four representative shower shape variables, comparing nominal (single-only) and mixed (22.4% double-blended) MC.

Table 13: Data–Inclusive MC χ^2/ndf for nominal and mixed MC at cut1, 0 mrad period. Lower values indicate better agreement.

Variable	E_T bin [GeV]	Nominal	Mixed	Improvement
w_η^{CoG}	10–14 (pt0)	47.4	29.1	1.6×
w_η^{CoG}	14–18 (pt1)	17.6	3.0	5.9×
w_η^{CoG}	22–28 (pt3)	16.9	0.75	22.5×
BDT score	14–18 (pt1)	12.2	1.6	7.5×
e_{t1} (inner ring)	14–18 (pt1)	9.3	1.9	4.8×
$E_{1\times1}/E_{3\times3}$	14–18 (pt1)	4.5	4.1	1.1×

5.3.1 w_η^{CoG} (Eta Cluster Width)

w_η^{CoG} is the variable most sensitive to double interactions: vertex shifts displace clusters along the beam axis, broadening the η profile. The improvement grows with E_T: at 22–28 GeV, χ^2/ndf drops from 16.9 to 0.75 (22.5×), indicating that the nominal MC's failure to model pileup-induced broadening becomes increasingly significant relative to statistical uncertainties.



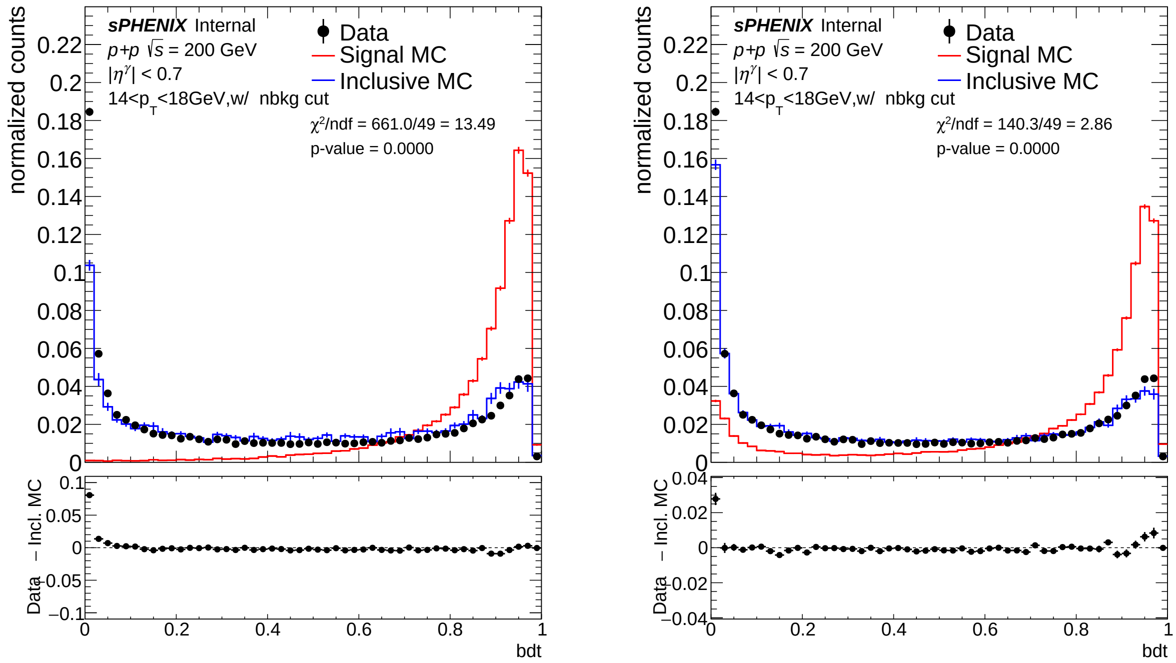
(a) Nominal MC

(b) Mixed MC

Figure 15: w_η^{CoG} at $14 < E_T < 18$ GeV, 0 mrad, cut1. Nominal χ^2/ndf = 17.6; mixed χ^2/ndf = 3.0 (5.9× improvement).

5.3.2 BDT Score

The BDT score amplifies underlying shower shape distortions through the XGBoost decision function, making it the most sensitive composite variable.



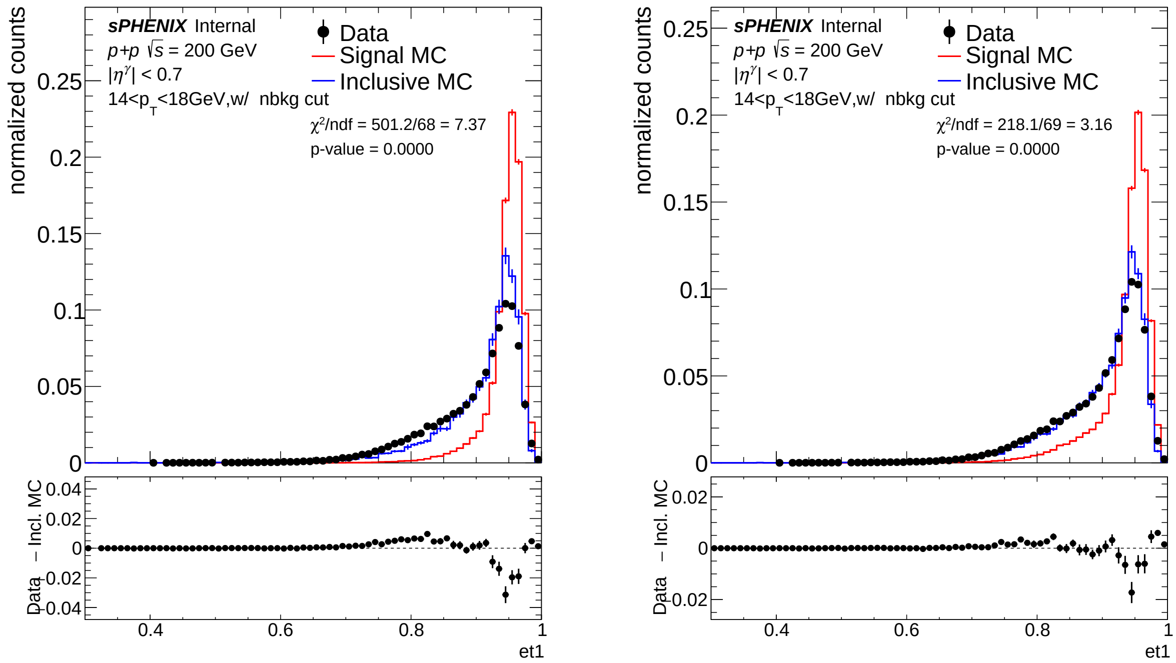
(a) Nominal MC

(b) Mixed MC

Figure 16: BDT score at $14 < E_{\mathrm{T}} < 18$ GeV, 0 mrad, cut1. Nominal $\chi^2/\mathrm{ndf} = 12.2$; mixed $\chi^2/\mathrm{ndf} = 1.6$ ($7.5\times$ improvement).

5.3.3 e_{t1} (Inner Ring Energy Fraction)

Double interactions spread energy outward from the shower core, reducing e_{t1}.

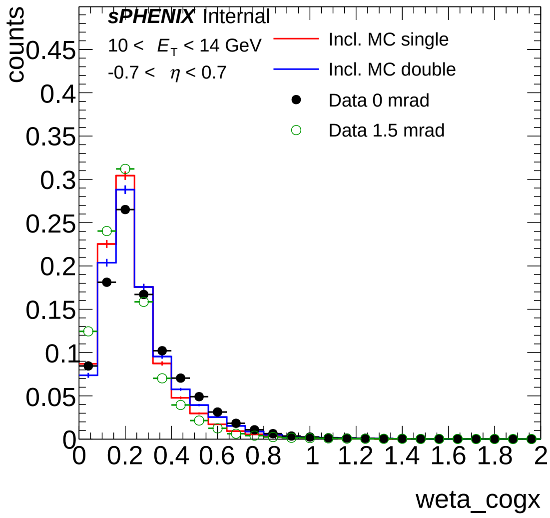


(a) Nominal MC (b) Mixed MC

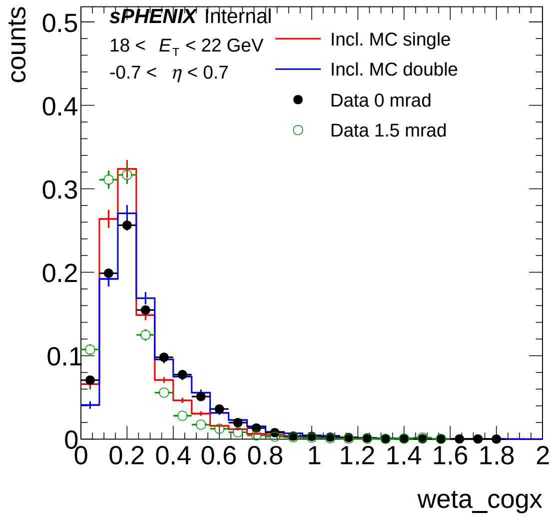
Figure 17: e_{t1} at $14 < E_{\mathrm{T}} < 18$ GeV, 0 mrad, cut1. Nominal $\chi^2/\mathrm{ndf} = 9.3$; mixed $\chi^2/\mathrm{ndf} = 1.9$ (4.8× improvement).

5.3.4 Single vs. Double Interaction Overlay

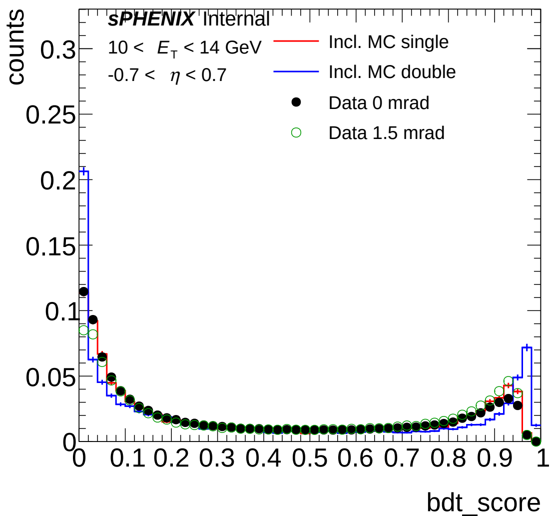
Figure 18 overlays single-interaction MC (red) and double-interaction MC (blue) with data from both crossing-angle periods. Data at 0 mrad systematically tracks the blended expectation (closer to the double MC tail), while data at 1.5 mrad tracks closer to single MC, consistent with the different pileup fractions.



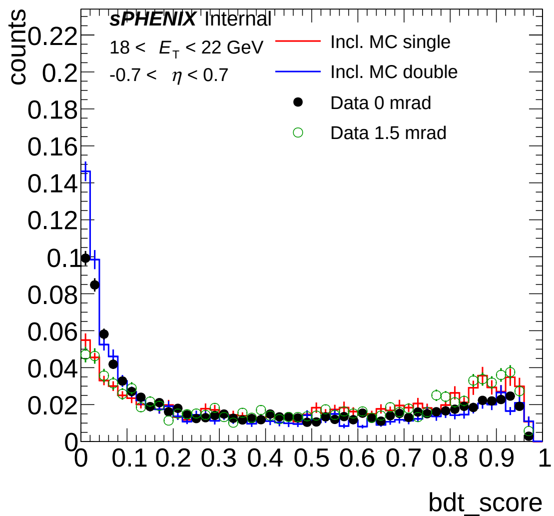
(a) w_η^{CoG}, 10–14 GeV



(b) w_η^{CoG}, 18–22 GeV



(c) BDT score, 10–14 GeV



(d) BDT score, 18–22 GeV

Figure 18: Shower shape distributions: single MC (red), double MC (blue), data 0 mrad (filled black), data 1.5 mrad (open green). Left: $10 < E_T < 14$ GeV. Right: $18 < E_T < 22$ GeV.

5.4 Results: 1.5 mrad Period

The 1.5 mrad period (nominal dataset) has a smaller double-interaction fraction (7.9% cluster-weighted), so the mixing effect is correspondingly smaller. The improvement pattern is consistent with 0 mrad but reduced in magnitude, confirming that the effect scales with the pileup fraction.

5.5 Discussion

The mixed MC dramatically improves data–MC agreement for pileup-sensitive variables (w_η^{CoG} and BDT score), with χ^2/ndf improvement factors of 1.6× to 22.5×. The improvement grows with E_T: at 22–28 GeV, the nominal MC discrepancy is most systematic relative to statistical uncertainties, and the mixing provides the largest relative correction.

The $E_{1\times1}/E_{3\times3}$ ratio shows only 1.1× improvement, indicating this variable is less sensitive to pileup-induced broadening or that competing effects partially cancel.

For the nominal cross-section measurement using the 1.5 mrad period, the pileup mixing is **not applied as a correction** because:

1. The pileup fraction is small (7.9% cluster-weighted).
2. The genuine pileup effects on efficiency (Table 11) amount to ∼9% relative for the mixed 0 mrad sample and less for 1.5 mrad, within the existing systematic uncertainty budget.
3. The effect is dominated by the reconstruction and BDT stages, which are partially correlated with vertex-shift systematics already evaluated.

The study serves as a cross-check that unmodeled pileup does not significantly bias the measurement.

6 Purity, Efficiency, and Closure

This section collects the key performance plots for the nominal analysis: data purity from the ABCD method, MC purity closure, the MC purity non-closure correction, and the factorized efficiency components.

6.1 Purity

Figure 19 shows the photon purity extracted from data using the ABCD double-sideband method, with and without signal leakage corrections (c_B, c_C, c_D), fitted by a Padé [1/1] rational function. Figure 20 shows the corresponding MC closure test, comparing the truth-level purity to the ABCD-estimated purity in inclusive MC.

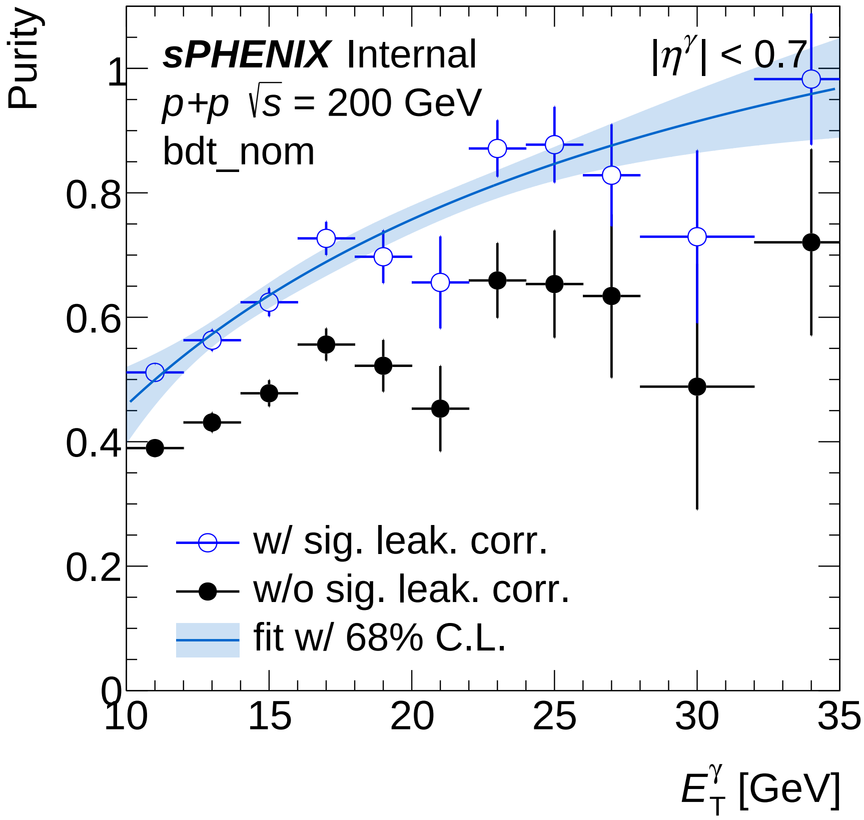


Figure 19: Photon purity as a function of E_{T} in data, extracted via the ABCD method with (solid) and without (open) signal leakage corrections, fitted by a Padé [1/1] rational function. The leakage correction reduces the purity estimate by 5–15% at low E_{T}.

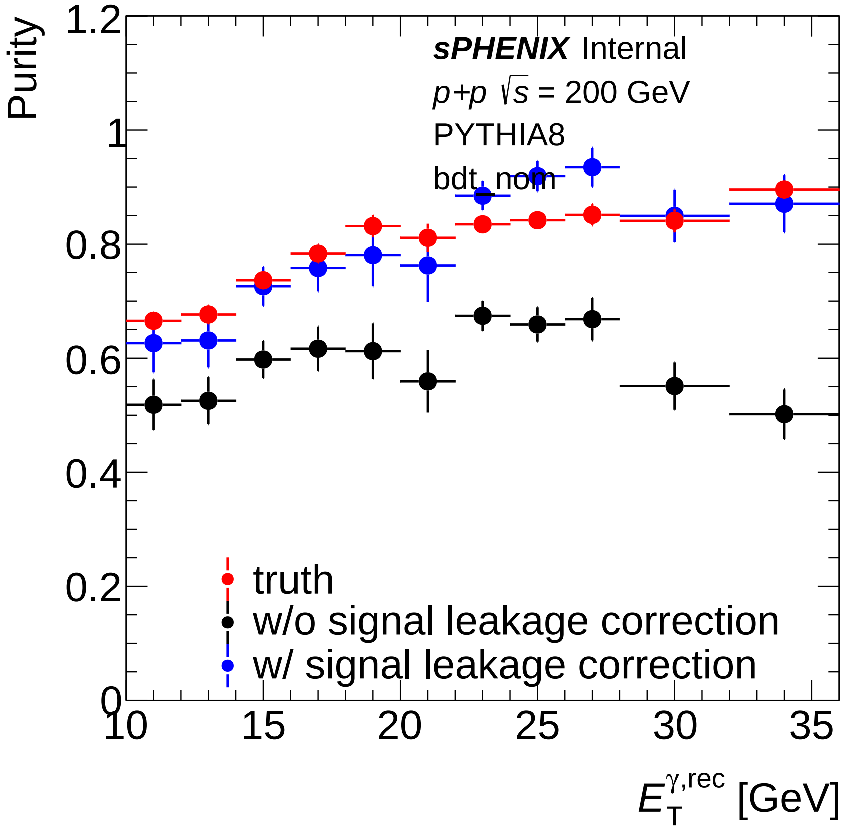


Figure 20: MC purity closure test: truth-level purity (black) vs. ABCD-estimated purity with (red) and without (blue) leakage corrections, in inclusive MC. The ABCD method reproduces the truth purity to within 5–10% at low E_{T}, with the residual non-closure taken as a systematic correction.

6.2 MC Purity Non-Closure Correction

The MC purity non-closure is defined as the ratio of ABCD-estimated purity to truth purity in MC, evaluated bin by bin in E_{T}. Figure 21 shows this correction factor across all E_{T} bins. Values above 1.0 indicate that the ABCD method overestimates the purity; values below 1.0 indicate underestimation. The correction is applied multiplicatively to the data purity before computing the cross-section, and its deviation from unity is taken as a systematic uncertainty.

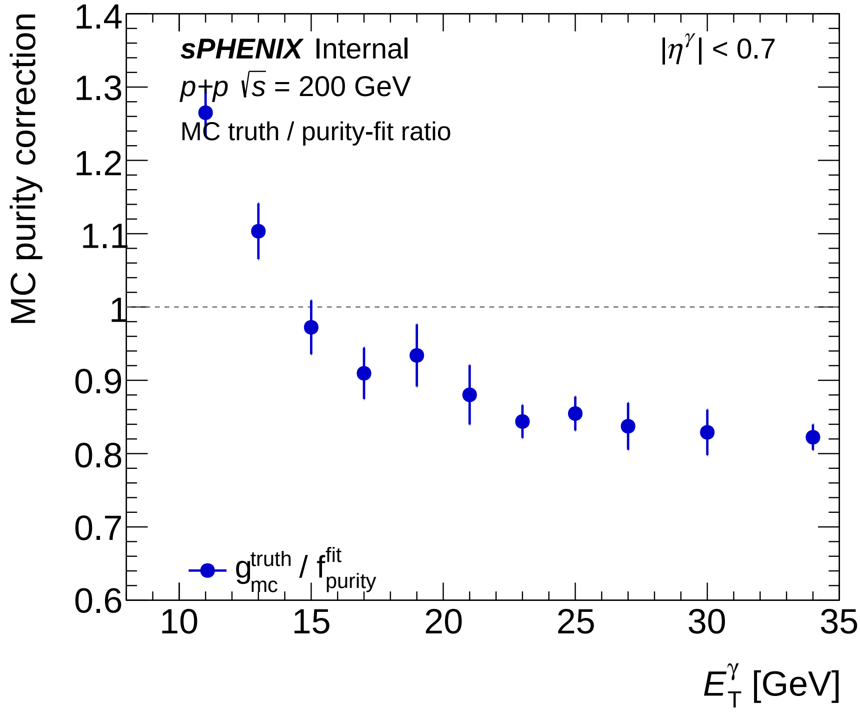
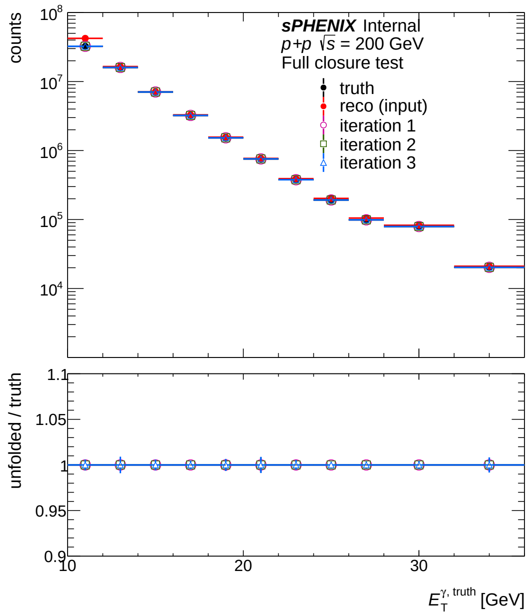


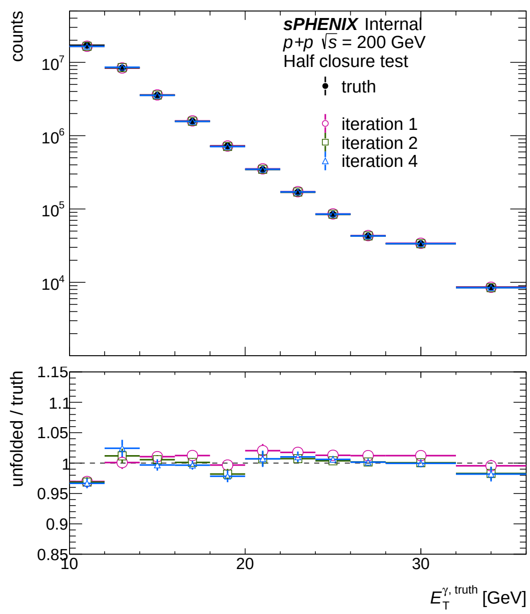
Figure 21: MC purity non-closure correction factor (ABCD-estimated / truth purity) as a function of E_{T}. The correction is typically 0.9–1.1, with larger deviations at low E_{T} where signal leakage corrections are most significant.

6.3 Unfolding Closure

Figures 22a and 22b show the unfolding closure tests using the full MC sample and an independent half-sample, respectively.



(a) Full closure (same MC for response + test)

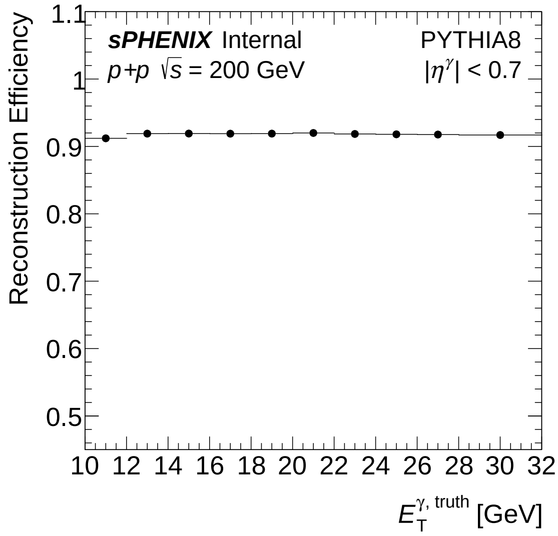


(b) Half closure (independent MC halves)

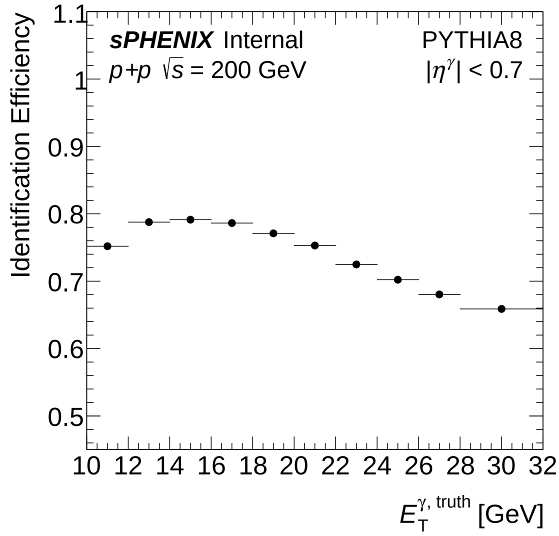
Figure 22: Unfolding closure tests. Left: full closure reproduces truth within ∼1%. Right: half closure (independent training and test samples) agrees within ∼2%, confirming the response matrix is not overfit.

6.4 Nominal Efficiency Components

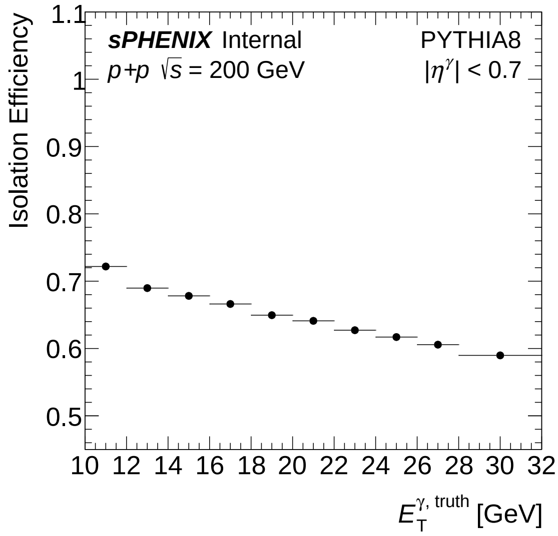
Figure 23 shows the four factorized efficiency components from single-interaction MC as used in the nominal cross-section extraction.



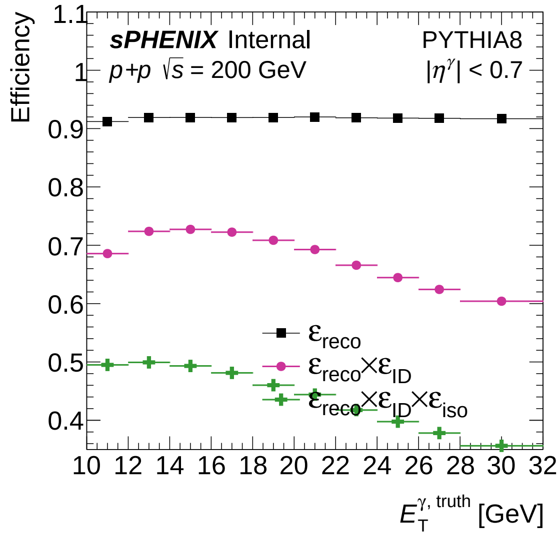
(a) Reconstruction efficiency



(b) Photon ID (BDT) efficiency



(c) Isolation efficiency



(d) Total (all combined)

Figure 23: Factorized photon efficiency components from single-interaction MC vs. truth E_{T} (nominal `base_v1E` model, `trkID`-only matching). Top left: reconstruction (93–95%). Top right: photon ID / BDT (81–92%). Bottom left: isolation (60–77%). Bottom right: total combined (52–58%).

7 Summary

Table 14 provides a concise overview of the four work streams and their impact on the cross-section measurement.

Table 14: Summary of updates since the preliminary result.

Work stream	Key change	Result	Cross-section impact
BDT photon ID	6 → 25 features, 11 model variants, E_{T}-dependent threshold	Systematic framework with 30+ configs; 5–10% uncertainty from threshold	Improved discrimination and well-characterized systematics
NPB rejection	Hard timing cut → 25-feature XGBoost score	∼99% purity at ∼98% signal retention	Cleaner data with flexible threshold systematics
Double interaction efficiency	ΔR removed; decomposition into genuine pileup effects	Reco −20 pp, BDT −13 pp, iso −3 pp in double MC	Single-MC efficiency used; ∼9% relative pileup effect at 0 mrad
Shower shape comparisons	Pileup-blended MC templates	χ^2/ndf improvement up to 22×	Cross-check confirms pileup within existing systematics

The combined conclusion is that:

1. The $\Delta R < 0.1$ truth-matching requirement was a vertex-resolution artifact, rejecting ∼3.5% of well-reconstructed photons even in single-interaction MC. It has been removed; track ID matching alone is definitive.
2. With ΔR removed, the genuine pileup effects are quantified: reconstruction degrades by 20 pp, BDT identification by 13 pp (from vertex-shifted E_{T} and shower shapes), and isolation by 3 pp in double MC.
3. The NPB rejection provides cleaner data samples with well-controlled systematics.
4. Pileup-blended MC templates substantially improve data–MC shower shape agreement (χ^2/ndf improvement up to 22×), validating the physics-motivated mixing fractions.
5. For the nominal 1.5 mrad measurement with 7.9% cluster-weighted pileup, the total efficiency impact is ∼a few percent relative and within existing systematic uncertainties.